The container Method

1) Mantel's theorem in random graphs:

$$\mathrm{ex}(G,H) = \max\left(|E(G')| : G' \subseteq G \text{ and } H \not\subseteq G \right\}.$$

Mantel theorem 1907 $\mathrm{ex}(K_n, \Delta) = \left\lfloor \frac{n^2}{4} \right\rfloor$

Turán's theorem 1940: $\mathrm{ex}(K_n, K_{r+1}) = \left(1 - \frac{1}{r} + o(1)\right) \frac{n^2}{2}$

The largest Δ-free graph is $K_{\frac{n}{2},\frac{n}{2}}$.

Question: What is $\mathrm{ex}(G(n,p), \Delta)$?

$\rightarrow$ Intersection with $K_{\frac{n}{2},\frac{n}{2}}$:

$$\mathrm{ex}(G(n,p), K_3) \geq \left(\frac{1}{4} + o(1)\right) p n^2$$

$\rightarrow$ Triangle removal process:

$$\text{if } \underbrace{\mathbb{E}[\#\text{ triangles}]}_{= p^3 \frac{n^3}{6}} \ll \underbrace{\mathbb{E}[|E(G(n,p))|]}_{= p n^2} \quad \text{i.e. } p \ll 1/\sqrt{n}$$

then we can remove the triangles one by one and

$$\mathrm{ex}(G(n,p), K_3) \geq p\binom{n}{2}(1 - o(1)) = \left(\frac{1}{2} + o(1)\right) p n^2$$

Frankl–Rödl 1986, Haxell–Kohayakawa–Łuczak 1996:

If $p \gg 1/\sqrt{n}$ $\mathrm{ex}(G(n,p), K_3) = \left(\frac{1}{4} + o(1)\right) p n^2$ w.h.p.

∘ Usual approach : T_m = # triangle-free subgraphs of $G(n,p)$ with m edges.

Show that $\mathbb{E}[T_m] = o(1)$ when $m > (1+o(1))\frac{n^2}{4}p$

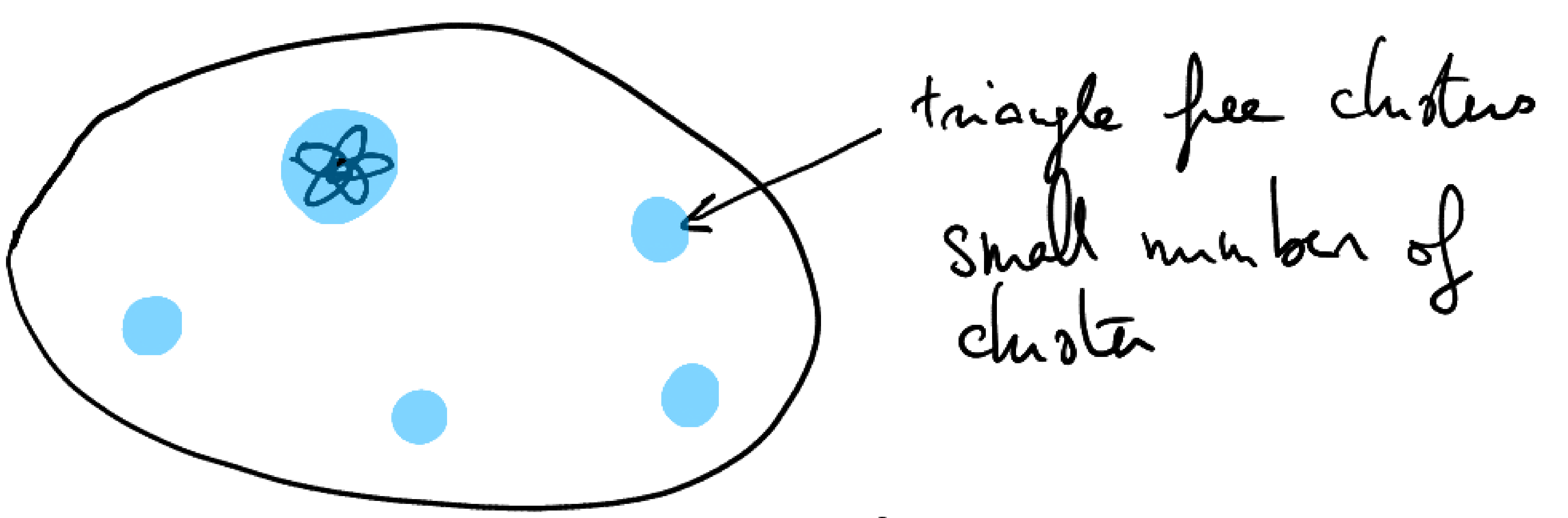
→ It's false. There are many triangle-free subgraphs of $G(n,p)$ with $m \approx p\frac{n^2}{4}$ edges :

$$\mathbb{E}[T_m] \geq \binom{ex(K_n, K_3)}{m} p^m = \binom{n^2/4}{m} p^m$$

$$\sim \left((1+o(1))\frac{epn^2}{4m}\right)^m \gg 1.$$

The number of big triangle-free graphs blows up but we might still have none of them w.h.p.

The triangle-free subgraphs are clustered together ⇝ their appearance are positively correlated events.



space of all subgraphs of $G(n,p)$ with Hamming distance

Container theorem for Δ-free graphs:

$\forall n > 0$, $\exists \mathcal{G}_n$ a collection of graphs on n vertices s.t.

(i) $|\mathcal{G}_n| \leq n^{O(n^{3/2})}$

(ii) Each $G \in \mathcal{G}_n$ has at most $o(n^3)$ triangles

(iii) Each triangle-free graph on n vertices is contained in some $G \in \mathcal{G}_n$

Supersaturation for triangles:

$\forall \varepsilon > 0$, $|E(G)| \geq (1+\varepsilon)\frac{n^2}{4} \implies G$ has $\Omega(n^3)$ triangles.

Theorem: $p \gg \log n / \sqrt{n} \implies ex(G(n,p), K_3) = \left(\frac{1}{4} + o(1)\right) p n^2$

Proof: Let $\varepsilon > 0$ and $m = \left(\frac{1}{4} + \varepsilon\right) p n^2$.

Goal: $\mathbb{P}(\exists H \subseteq G(n,p)$ with $e(H) = m$, no triangles$) = o(1)$

By supersaturation, for every $G \in \mathcal{G}_n$, $e(G) \leq \left(\frac{1}{4} + o(1)\right) n^2$

Let X_G be the number of edges of $G \cap G(n,p)$.

$X_G \sim Bin(e(G), p)$

$$\mathbb{P}(X_G \geq m) \leq e^{-\Omega(\varepsilon p n^2)} \quad \text{By Chernoff bound}$$

Let Y_m be the number of graphs of $\mathcal{G}_n$ with at least m edges in $G(n,p)$.

$$\mathbb{E}[Y_m] \underset{\text{union bound}}{\leq} \sum_{G \in \mathcal{G}_n} \mathbb{P}(X_G \geq m)$$

$$\leq n^{O(n^{3/2})} e^{-\Omega(\varepsilon p n^2)} = o(1) \quad \text{if } p \gg \frac{\log n}{\sqrt{n}}$$

$\blacksquare$

2) A container lemma for hypergraphs.

Encoding triangles of K_n with a hypergraph:

Let $\mathcal{H}$ be the 3-uniform hypergraph whose vertices are the edges of K_n and whose hyperedges are the triples of edges forming a triangle.

A triangle free graph on n-vertices corresponds to an independent set of K_n.

Hypergraph container lemma for 3-uniform hypergraphs:

For every $c>0$, there exists $\delta>0$ such that the following holds. For every 3-uniform hypergraph $\mathcal{H}$ of average degree d, such that $\Delta_1(\mathcal{H}) \leq c \cdot d$ and $\Delta_2(\mathcal{H}) \leq c\sqrt{d}$, there exists a collection $\mathcal{C}$ of subsets of $V(\mathcal{H})$ and a function $f: \{0,1\}^{V(\mathcal{H})} \to \mathcal{C}$ such that

(i) For every independent set I of $\mathcal{H}$, there exists a fingerprint S of I with $|S| \leq v(\mathcal{H})/\sqrt{d}$ and
$$S \subseteq I \subseteq f(S)$$

(ii) $|C| \leq (1-\delta)\, v(\mathcal{H})$ for every $C \in \mathcal{C}$.

Observation: • We recall that $\Delta_k(\mathcal{H}) = \max_{\substack{|S|=k \\ S \subseteq V(\mathcal{H})}} |\{e \in E(\mathcal{H}) : S \subseteq e\}|$

• This implies the weak container theorem for triangle free graphs.

Let $\mathcal{H}$ be the 3-uniform hypergraph encoding the triangles of K_n.

We have $\Delta_1(\mathcal{H}) = n-2$ and $\Delta_2(\mathcal{H}) = 1$. Take $c=1$.

$\mathcal{C}$ has size at most $\binom{v(\mathcal{H})}{v(\mathcal{H})/\sqrt{d}} = \binom{n^2/2}{n^{3/2}/2} = n^{O(n^{3/2})}$

(i) $|\mathcal{G}| \in n^{O(n^{3/2})}$

(ii') $\forall S \in \mathcal{G}_{n},\ e(S) \leq (1-\delta)\, e(\mathcal{H}) = (1-\delta)\binom{n}{2}$

(iii) every triangle free graph is contained in some container of $\mathcal{G}$.

How do we get (ii) from (ii')?

For every fingerprint S such that $f(S)$ has more than $o(n^3)$ triangles, we apply the container lemma to $\mathcal{H}[f(S)]$.

$$e(\mathcal{H}[f(S)]) \geq \varepsilon n^3 \geq \varepsilon e(\mathcal{H})$$

so $\operatorname{avgdegree}(\mathcal{H}[f(S)]) = d' \leq \varepsilon d$ $\quad \Delta_1(\mathcal{H}[f(S)]) \leq \Delta_1(\mathcal{H}) \leq \frac{1}{\varepsilon} d'$

$$\Delta_2(\mathcal{H}[f(S)]) = 1.$$

so the conditions are still verified, with $c = \frac{1}{\varepsilon}$.

If we iterate this inductively t times, we get a collection $\mathcal{G}$ of graphs on n vertices, and a function $f : \{0,1\}^n \to \mathcal{G}$:

(i) $|\mathcal{G}| \leq \left(n^{O(n^{3/2})}\right)^t$

(ii) each $S \in \mathcal{G}$ has either $(1-\delta)^t \binom{n}{2}$ edges
or $\leq \varepsilon n^3$ triangles

(iii) every triangle-free graph H on n vertices has a fingerprint S s.t. $S \subseteq H \subseteq f(S)$, $|S| = O(n^{3/2})$

For t sufficiently large constant (depending on δ, which depends on c, which depends on ε)

$(1-\delta)^t \leq \varepsilon$ and so the number of triangles in each container $f(S)$ is at most $e(G)\cdot(n-2) \leq \varepsilon n^3$.

3) Proof of Frankl - Rödl, without the log factor:

Let $\varepsilon > 0$. Let $m = \left(\frac{1}{4}+\varepsilon\right)pn^2$.

For every fingerprint S, the probability that $G(n,p)$ contains a triangle-free graph H with m edges and fingerprint S is at most

$$\mathbb{P}(S \subseteq G(n,p)) \cdot \mathbb{P}\Big(e\big(f(S) \cap G(n,p)\big) \geq m - e(S)\Big)$$

As $f(S)$ contains at most $o(n^3)$ triangles, $f(S)$ has at most $\left(\frac{1}{4}+\varepsilon\right)n^2$ edges by supersaturation.

so $\mathbb{P}\Big(e\big(f(S)\cap G(n,p)\big) \geq m - e(S)\Big) \leq e^{-\Omega(pn^2)}$

because $p \gg n^{-1/2}$, so $e(S) \ll m$.

So by union bound, $\mathbb{P}(G(n,p)$ has a triangle free graph on m edges$)$

$$\leq \sum_{S \in \mathcal{C}} p^{e(S)} e^{-\Omega(pn^2)} \leq \sum_{k=0}^{O(n^{3/2})} p^k \binom{n^2/2}{k} e^{-\Omega(pn^2)}$$

$$\leq \sum_{k}^{O(n^{3/2})} \left(\frac{O(pn^2)}{k}\right)^k e^{-\Omega(pn^2)}$$

$$= o(1)$$

4) Other applications of the container method

Let $[n]_p$ denote the random subset of $[n]$ containing each element independently with probability p.

Supersaturation theorem for 3-arithmetic progressions:
For every $\varepsilon > 0$, there exists $\gamma > 0$ s.t. every subset $A \subseteq [n]$ of size εn contains at least γn^2 triples $\{a, b, c\}$ with $a - b = b - c$.

Theorem: For every $\varepsilon > 0$, there exists $C > 0$ s.t. $\forall p \geq C n^{-1/2}$, every subset A of $[n]_p$ of size at least $\varepsilon p n$ contains an arithmetic progression of length 3.

Proof:
Step 1: Try the first moment method

Step 2: Prove a container lemma.

1. The 3-arithmetic progressions can be encoded by a 3-uniform hypergraph $\mathcal{H}$ whose vertex set is $[n]$, with an edge for every triple $\{a, b, c\}$ s.t. $a - b = b - c$.

2. What can we say about $\mathcal{H}$?

- $e(\mathcal{H}) = \sum_{k=1}^{\lfloor n/2 \rfloor} n - 2k \leq \frac{n^2}{2} - 2 \cdot \frac{\left(\frac{n}{2}+1\right)}{2} \frac{n}{2} \approx \frac{n^2}{4}$.

  so $\mathcal{H}$ has average degree $d \approx \frac{n}{4}$.

- $\Delta_1(\mathcal{H}) \leq \frac{3n}{2} \leq 6d$ and $\Delta_2(\mathcal{H}) \leq 3$

- For every induced subgraph $\mathcal{H}'$ of $\mathcal{H}$ with at least γn^2 edges, where $d(\mathcal{H}') =$ avg degree $(\mathcal{H}') = \frac{e(\mathcal{H}')}{v(\mathcal{H}')} \geq \frac{e(\mathcal{H}')}{v(\mathcal{H})} \geq \gamma n$

and $\Delta_1(\mathcal{H}') \leq \Delta_1(\mathcal{H}) = \frac{3n}{2} \leq \frac{3}{2\gamma} \; d(\mathcal{H}')$

So the 3-uniform hypergraph container lemma can be applied to every induced subgraph $\mathcal{H}'$ of $\mathcal{H}$ with at least γn^2 hyperedges, taking $c = \frac{3}{2\gamma}$.

3. A container lemma for 3-arithmetic progressions.

For every γ, there exists $\delta > 0$ such that the following holds for every n. There exists a collection $\mathcal{C}$ of subsets of $[n]$ and a function $f : 2^{[n]} \to \mathcal{C}$ such that:

(i) every subset $I \subseteq [n]$ without 3-arithmetic progression has a fingerprint S s.t. $S \subseteq I \subseteq f(S)$ and $|S| \leq O(n^{1/2})$

(ii) every $C \subseteq \mathcal{C}$ has at most γn^2 arithmetic progressions of length 3.

Proof: We apply inductively apply the container lemma to $\mathcal{H}$ and its containers having more than γn^2 3-AP, taking $c = \frac{3}{2\gamma}$.

If we do this t times, each C in $\mathcal{C}$ has size at most $(1-\delta(c))^t \cdot n \leq 2\sqrt{\gamma}\, n$ for t a large constant.

so $e(\mathcal{H}[C]) \leq \frac{v(\mathcal{H}[C])^2}{4} \leq \gamma n^2$.

■

4. Take $\gamma(\varepsilon/2)$ from the supersaturation theorem.
For each fingerprint S, the probability $[n]_p$ contains a subset I of size $\varepsilon p n$, with no 3-AP and $S \subseteq I$ is $\leq$

$P(S \subseteq [n]_p) \cdot P(|[n]_p \cap f(S) \setminus S| \geq \varepsilon p n - |S|)$,

- $\mathbb{P}(S \subseteq [n]_p) = p^{|S|}$.
- $f(S)$ has at most δn^2 3-AP so by supersaturation, $|f(S)| \leq \frac{\varepsilon n}{2}$.

so $\mathbb{P}\left(\left|[n]_p \cap f(S) \setminus S\right| \geq \varepsilon n p - |S|\right) \leq \mathbb{P}\left(\left|[n]_p \cap f(S)\right| \geq \frac{3\varepsilon}{4} np\right)$
$\leq e^{-\Omega(\varepsilon n p)}$ by Chernoff.

So $\mathbb{P}(\exists A \subseteq [n]_p$ with no 3AP and $|A| = \varepsilon n p) \leq \sum_{k=1}^{2\delta n^{1/2}} \binom{n}{k} p^k e^{-\Omega(\varepsilon n p)}$

$\leq \sum_{k}^{2\delta n^{1/2}} \left(\frac{C\sqrt{n}}{k}\right)^k e^{-\Omega(\varepsilon n p)}$

because $p = Cn^{1/2}$.

$= o(1)$

for C large enough. ∎

<u>Proof</u> of supersaturation for 3-AP:

Let X, Y, Z be uniform independent random elements of A.

$$\mathbb{P}(X - Y = Y - Z) = \sum_{x, y \in A} \mathbb{P}(X = x)\, \mathbb{P}(Y = y)\, \mathbb{P}(Z = 2y - x)$$

$=$

5) Proof of the strong container lemma:

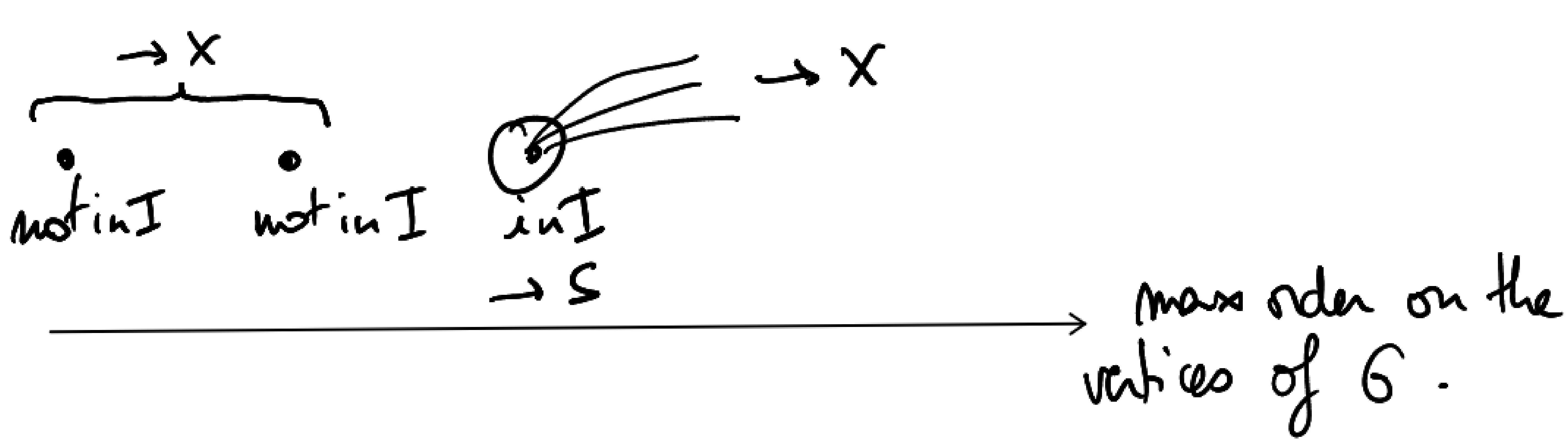
For simplicity, we prove a graph version: (Kleitman - Winston 80)

Theorem: For every $c > 0$, there exists $\delta > 0$ such that:
For every graph G with average degree d and $\Delta(G) \leq c \cdot d$, there exists a collection $\mathcal{C}$ of subsets of $V(G)$ and a function $f : \mathcal{I}(G) \to \mathcal{C}$ s.t.

(i) For every independent set I of G, there exists a fingerprint S of I with $|S| \leq 2\delta v(G)/d$ and
$$S \subseteq I \subseteq f(S)$$

(ii) $|C| \leq (1-\delta) v(G)$ for every $C \in \mathcal{C}$.

Proof: We maintain a partition $V(G) = A \sqcup S \sqcup X$ where A are the active vertices, S is the current fingerprint of I and X are the excluded vertices.
At first $V(G) = A$, $S = X = \emptyset$.



While $|X| \leq \delta v(G)$,
1. Let v be the first vertex in the max order of A, that belongs to I.
2. Move v from A to S.
3. Move the vertices before v and all the neighbours of v from A to X

- If I and I have the same fingerprint then, this means the algorithm ran identically, so $X(I) = X(I')$ and $A(I) = A(I')$.

so we set $f(S) = V(G) \setminus X$ which proves (ii)

- We need to prove $\forall I$, $|S(I)| \leq \tau v(G)$ where $\tau = 2\delta/d$

To do so, we prove that each time we add a vertex to S, we add at least $d/2$ vertices to X.

If at step i, $|S_i| \leq \tau v(G)$, $|X| \leq \delta v(G)$ and $W_i \{u \in A_i : u < v_i$ in the max order on $G[A_i]\}$ has size at most $\frac{d}{2}$,

$$e(G[A_i \setminus W_i]) \geq e(G) - \left(|X_i| + |S_i| + |W_i|\right) \cdot \Delta(G)$$

$$\geq e(G) - ((\tau+\delta)v(G) + d/2)\Delta(G)$$

$$\geq e(G) - \left(3\delta v(G) + d/2\right) c d$$

otherwise $|S| = 1$.

$$\geq e(G) - 4\delta v(G) c d$$

$d \leq 2\delta v(G)$

$$\geq (1 - 2\delta c) e(G) \geq \frac{e(G)}{2}$$ if δ small enough

So v has maximum degree at least $d/2$ in $G[A_i \setminus W_i]$ and all its neighbours are added to X.

Idea for 3-uniform hypergraphs: (Saxton-Thomason and Balogh-Morris-Samotij 15)

We maintain a graph G of constraints.
Each time we add a vertex v to S, we add all the edges uw such that $uvw \in E(\mathcal{H})$.

. the average degree of G is controlled because at each step, we add $\approx d$ edges to G. So after $\approx \sqrt{d}$ steps $\text{avgdg}(G) \approx \sqrt{d}$.

I is also an independent set of G so we can apply the container lemma to G. To do so, we need to control the max degree of G, we stop adding edges when a vertex has at least $c\sqrt{d}$ neighbours in G.

$\rightarrow \left| \begin{array}{l} \Delta_1 \\ \Delta_2 \end{array} \right.$ this does not cost us to many edges

6) Ramsey for random graphs

Every 2-colouring of K_6 has a monochromatic triangle. Can we construct a K_4-free graph with this property?

Frankl–Rödl 1986 & Rödl–Ruciński 1994:
For every $r \in \mathbb{N}^*$ there exists $C > 0$ s.t. for $p \gg C n^{-1/2}$, then with probability $1 - o(1)$, every r-colouring of the edges of $G(n,p)$ has a monochromatic triangle.

Proof (by Nenadov – Steger 2016, with $\log n$ instead of C, for simplicity)

Supersaturation for Ramsey: For every $r \in \mathbb{N}^*$, there exists $\varepsilon > 0$ such that for all sufficiently large n, every $(r+1)$-colouring of the edges of K_n contains at least $(r+1)\varepsilon n^3$ monochromatic triangles.

We view a r-colouring of $G(n,p)$ as a partition of the edges of $G(n,p)$ into r graphs $(G_1, \dots G_r)$ i.e.
$E(G(n,p)) = \bigsqcup_{i \in [r]} E(G_i)$.

$G(n,p)$ admits a r-colouring without monochromatic triangle if there exists $H_1, \dots H_r$ triangle-free $\subseteq G(n,p)$ s.t. $G(n,p) = H_1 \sqcup \dots \sqcup H_r$.

So we use the container theorem for triangle-free graphs.

$\mathbb{P}(G(n,p)$ admits a bad colouring$)$

$$= \mathbb{P}(\exists H_1, \dots H_r \text{ triangle-free s.t. } G(n,p) \subseteq H_1 \sqcup \dots H_r)$$

union bound over r-tuples of containers
$$\leq \sum_{G_1, \dots G_r \in \mathcal{G}} \mathbb{P}(G(n,p) \subseteq G_1 \cup \dots G_r)$$

Each G_i has at most εn^3 triangles so by supersaturation, $K_n \setminus (G_1 \cup \dots G_r)$ has at least εn^3 triangles. As each edge of $K_n \setminus (G_1 \cup \dots G_r)$ is contained in at most n triangles,

$$e(K_n \setminus (G_1 \cup \dots G_r)) \geqslant \varepsilon n^2 .$$

So $$P(G(n,p) \subseteq G_1 \cup \dots G_r) = (1-p)^{e(K_n \setminus (G_1 \cup \dots G_r))}$$

$$\leq (1-p)^{\varepsilon n^2} \leq e^{-\varepsilon p n^2}$$

hence $$P(G(n,p) \text{ admits a bad colouring}) \leq \left(n^{O(n^{3/2})}\right)^r e^{-\varepsilon p n^2} \longrightarrow 0$$

$$\text{if } p \gg \frac{\log n}{\sqrt{n}} . \quad \blacksquare$$

Using the k-uniform hypergraph container lemma:

[Conlon - Gowers 2016 & Schacht 2016) $\forall \varepsilon > 0$, $\forall k \in \mathbb{N}$, there exists C s.t. for $p \geq C n^{-1/(k-1)}$, with probability $1-o(1)$, every subset $A \subseteq [n]$ of size at least $\varepsilon p n$ contains an arithmetic progression of length k.

There are many other beautiful lications of this method. For example

- Discrete geometry (see exercise sheet)
- Counting H-free graphs.
- Additive combinatorics (sum free sets, Sidon sets ...)